



# A Dialectal Speech Corpus and Automatic Speech Recognition Models for three Jordanian Sub-Dialects

Final Project Report Submitted to  
The Department of Computer Science  
Faculty of Computer and Information Technology  
Jordan University of Science and Technology  
In Partial Fulfillment of the Requirements for the Degree of  
Bachelors of Science in Computer Science

Prepared by:

Majd Fakhri Omari [163099]  
Rania Khalil Ali Hushoush [163339]

Supervisor:  
Dr. Rasha Obeidat

January 2026

## نموذج حقوق الملكية الفكرية لمشاريع التخرج في قسم علوم الحاسوب

يتم قراءة وتوقيع هذا النموذج من قبل الطلاب المسجلين لمشاريع التخرج في قسم علوم الحاسوب

تعود حقوق الملكية الفكرية لمشاريع التخرج وتنتجها (مثل براءات الاختراع أو أي منتج قابل للتسويق) إلى جامعة العلوم والتكنولوجيا الأردنية، وتخضع هذه الحقوق إلى قوانين وأنظمة و تعليمات الجامعة المتعلقة بالملكية الفكرية وبراءات الاختراع.

بناء على ما سبق أوافق على ما يلي:

- (1) أن أحفظ كافة حقوق الملكية الفكرية لجامعة العلوم والتكنولوجيا الأردنية في مشروع التخرج.
- (2) أن ألتزم بوضع اسم جامعة العلوم والتكنولوجيا الأردنية و أسماء جميع الباحثين المشاركين في المشروع على أي نشرة علمية للمشروع كاملا أو لنتائجه. و يشمل ذلك النشر في المجلات و المؤتمرات العلمية عامة أو النشر على المواقع الإلكترونية أو براءات الاختراع أو المسابقات العلمية.
- (3) أن ألتزم بأسس حقوق التأليف المعتمدة في جامعة العلوم والتكنولوجيا الأردنية.
- (4) أن أقوم بإعلام الجهة المختصة في الجامعة عن أي اختراع أو اكتشاف قد ينتج عن هذا المشروع و أن ألتزم السرية التامة في ذلك و أن أعمل من خلال الجامعة على الحصول على براءة الاختراع التي قد تنتج عن هذا المشروع.
- (5) أن تكون جامعة العلوم والتكنولوجيا الأردنية هي المالك لأي براءة اختراع قد تنتج عن هذا المشروع و تشمل هذه الملكية حق الجامعة في إعطاء التراخيص و التسويق و البيع كمؤسسة راعية و داعمة لكافة الأنشطة البحثية. ويكون حق الطالب شمول اسمه على براءة الاختراع كأحد المخترعين، و في حال تم إعطاء تراخيص أو تسويق و بيع لأي من منتجات المشروع يمنح المخترعون بما فيهم الطالب نسبة من الإيرادات حسب تعليمات البحث العلمي في جامعة العلوم والتكنولوجيا الأردنية.

التوقيع

مجد فخري العمري (اسم الطالب)

التوقيع

رانيا خليل العشوش (اسم الطالب)

التوقيع

(اسم الطالب)

التوقيع

د. رشا عبيدات (اسم المشرع)

تاريخ 2026/6/9

# A Dialectal Speech Corpus and Automatic Speech Recognition Models for three Jordanian Sub-Dialects

**Abstract**—Many Jordanian Arabic dialects are still inadequately researched in speech recognition, which limits the capabilities of modern systems for Jordanian speech. Most current systems are designed to work with Modern Standard Arabic or general Jordanian dialects, reflecting a limited ability to understand Jordanian speech. This research aims to address this gap by developing and evaluating dialect-specific ASR systems for Jordanian Arabic. The research focuses on the collection of speech data from three different locations in the state of Jordan: Irbid in the north of the state, Amman as the capital of the state, and Northern Aghwar. The resulting corpus consists of approximately 4,500 speech recordings, corresponding to nearly 1.9 hours of speech data. The texts were originally written in the local dialect of the state of Jordan. The texts were then recorded in audio format to represent the pronunciation of the spoken words. The collected speech data was then used to train various speech recognition systems. The evaluated systems included several variants of the Whisper model and an Arabic Wav2Vec 2.0 model. Experimental results showed that dialect-specific fine-tuning substantially improved recognition accuracy across all evaluated dialects. Larger Whisper models consistently achieved lower WER and CER values, with Whisper Large obtaining the best overall performance. Cross-dialect evaluation experiments further showed that models trained on one Jordanian dialect were able to recognize speech from other Jordanian dialects, although recognition accuracy varied depending on the degree of phonetic and linguistic similarity between dialects. Finally, the paper highlights the need to use localized dialect data to improve the system’s accuracy.

## I. PROJECT GOALS AND OBJECTIVES

The primary goal of this project is to enhance voice recognition systems’ comprehension of Jordanian dialects by utilizing data that accurately represents regional speech patterns.

The specific objectives of this project are as follows:

- Collect speech data from three important parts of Jordan: the Northern Aghwar, Irbid, and Amman. Ensure the data reflects regional pronunciation or style.
- Before making an audio recording, translate written Arabic sentences into a real, daily Jordanian dialect.
- Use this dialectal speech to refine Arabic-focused Wav2Vec 2.0 models and assess their performance.
- Use WER and CER to investigate how different training conditions and hyperparameters impact model quality.

- To determine the comparative advantages of dialect-aware models and a general-purpose ASR system, compare them.
- To Examine mistakes qualitatively to identify the phonological and dialectal difficulties these models face.

## II. INTRODUCTION

Arabic Automatic Speech Recognition (ASR) has remained a challenge due to the great diversity of spoken dialects and limited speech resources. Recent large-scale efforts, such as the Casablanca Corpus [1], have expanded dialectal representation by providing multi-dialectal speech in Arabic across several Arabic-speaking countries, including Jordanian Arabic.

In this regard, speech-based research usually depicts Jordanian Arabic at a general dialectal level, either as a single category or as a component of larger Levantine groupings [2]. Although Jordanian-spoken audio is available in the current resources, the granularity of these datasets is typically low, making it challenging to capture variation in Jordanian speech.

Dialectal variation in Jordanian Arabic is strongly influenced by both geographic and social factors. In regions with stable communities and little population movement, local dialects are usually well-preserved and deeply embedded. On the other hand, governorates that attract migrants for work, education, or urban living often bring together speakers from diverse linguistic backgrounds. Instead of a single, stable local dialect, this interaction results in the emergence of mixed or fluctuating speech patterns

Inspired by this reality, this study highlights the importance of not treating the Jordanian dialect as a monolithic, unified entity. Indeed, it has become essential to understand the dialect of each governorate, especially in areas where demographic changes directly impact everyday speech. The key factor here is understanding the dialect used by people in each governorate in their everyday speech, especially in areas where population shifts influence daily speech patterns. A review of the existing literature indicates that previous studies—both in Arabic contexts and in other languages—generally classify dialects into broad regional categories without considering internal variations within these broad regions and overlooking the impact of population movements on

dialectal evolution. Furthermore, research on automatic speech recognition in Arabic has historically focused on Modern Standard Arabic or on a limited set of dialects, resulting in the underrepresentation of many dialects with limited resources, including Jordanian dialects, within broad regional categories.

This study presents governorate-level analysis of Jordanian Arabic for the purposes of automatic speech recognition, with a focus on speech from the following three regions: Irbid, Amman, and Northern Aghwar. The proposed dialect-aware data collection pipeline adapts written sentences into local Jordanian dialect before recording them as spoken audio, preserving natural pronunciation and Dialect . Using the resulting speech-text dataset, several ASR models were evaluated, including multiple Whisper model variants and an Arabic Wav2Vec 2.0 model. Model performance is assessed using Word Error Rate (WER) and Character Error Rate (CER), enabling a comparative analysis of how effectively each model captures dialectal variation across different Jordanian regions. In addition, cross-dialect evaluation experiments were conducted to investigate the ability of dialect-specific models to generalize across other Jordanian dialects.

### III. RELATED WORK

#### *A. The Multi-Dialectal Dilemma: A Global Technical Frontier in ASR*

Arabic language is among one of the largest linguistic foundations of humanity and serves as one of the major communication mediums for approximately 400 million people and a vital bridge between civilizations. As one of the six official languages of the United Nations [3]. The complexity of Arabic is attributed not only to its vast vocabulary but also to the productive morpho-syntactic style of the "root and pattern" system, which presents an inherent challenge for (NLP) and Speech Recognition [4].

The Arabic language is not only complex, but also exhibits diglossia. As reported in linguistic studies, the formal language of instruction (Modern Standard Arabic or MSA) differs significantly from many dialects [5]. Since their origins, some of these dialects have evolved into forms that have resulted in different systems with distinct pronunciations and vocabulary, making it difficult even for standard AI models to connect formal learning with conversation in the natural world.

The pursuit of the "authentic voice" of speakers is not only a local challenge for Arabic, but also part of an international technological struggle. We can see its magnitude by looking at other leading languages that have suffered from this fragmentation. English is an excellent example, where dialectal variations in its grammatical structure are also viewed as "moving targets" that change across space and time [6]. Elsewhere in the Far East, the Chinese linguistic landscape faces an even greater challenge with regard to "phonetic interference," where

dialects share a single writing system but differ radically in their phonetic pronunciation, a challenge that is a major challenge for speech recognition models [7].

The "rhythmic melody" of speech is also a problem. Complex deep learning models are required to identify such subtle musical cues in Korean dialects, where the true identity of the speaker in terms of his region typically lies in the intonation pattern rather than the words [8]. the concept of information theory has also been used as an analytical tool to quantify the distance between European and Brazilian Portuguese, showing how the commonality among two closely-related forms can make them indistinguishable with most AI systems. [9].

Studying Arabic from a global perspective reveals a global linguistic obstacle: the "problem of dialects". The vast scope and complexity of the Arabic language [10] underscore the need for a system that transcends "standard" models to encompass the rich, diverse, and natural variations of human dialects.

#### *B. Evolution of Arabic ASR: From Traditional to Neural Approaches*

The Arabic ASR platform has changed considerably in the last decade, mainly due to the scarcity of labeled data and the linguistic complexity of Arabic. Existing work covers classical methods of acoustic modeling, various data augmentation approaches, and the recent move towards self-supervised learning.

Menacer et al. [11], who used a GMM-HMM hybrid phonological model enhanced with a fine-tuned deep neural network (DNN) for Modern Standard Arabic (MSA). This study identified key challenges such as morphological richness and pronunciation ambiguity resulting from lack of diacritics. It was necessary to find a practical solution to address the limitations imposed on Arabic speech corpora and data augmentation. In [12], techniques such as noise injection, speed variation, and pitch variation were applied to the SASSC dataset, achieving a 4.55% reduction in the word error rate. In a recent study, the ArabRecognizer system [13] presented architectures inspired by DeepSpeech2, using stacked GRU layers with CTC decoding to improve accuracy and computational efficiency, especially in Arabic phonetics.

#### *C. Self-Supervised Learning and Multidialectal Advancements*

Self-supervised learning (SSL) has made significant progress in automatic speech recognition for resource-limited Arabic. Research conducted on the Wav2Vec 2.0 framework [14] has shown that multilingual models such as XLSR-53 significantly outperform English-only models when fine-tuned with Arabic data. This progress reached a new level with the release of XLS-R [15], a model pre-trained on 436,000 hours of speech, providing a robust global foundation for dialects with limited resources.

Since then, the focus has been on capturing the rich diversity of dialects across different regions. Among these works is the MGB-5 Challenge [16], which used 2,000 hours of media broadcasting to demonstrate that standardized models trained on multiple dialects achieve better generalization than those limited to a single dialect. Based on this, Shon et al. [17] laid the groundwork for dialect identification using convolutional neural network-based encoders. The Casablanca Project [1] recently presented a massive, community-driven dataset covering eight dialects, including the Jordanian dialect. Their findings confirmed that while single-dialect models fail on unseen varieties, **unified multidialectal training**—such as **Whisper-Mixed**—is essential for robust performance.

In addition to dialectal diversity, multilingual complexities and post-processing are important research areas. Chowdhury et al. [18] proposed Conformer-based models to support dialectal code-switching between Arabic, English, and French. Meanwhile, developments such as FastConformer have propelled the precision of Modern Standard Arabic (MSA) and Classical Arabic (CA) to state-of-the-art levels [19]. To improve these outputs, the CoVoGER benchmark has introduced generative error correction (GER) using Large Language Models (LLMs) to contextually correct recognition errors. [20]

Safieh et al. [21], presented the most relevant work to the current research, using Wav2Vec 2.0 technology and five-generation iterative training processes, along with advanced data enhancement techniques such as pitch variation and time dilation, on a 113-hour Jordanian dataset encompassing urban, rural, and Bedouin dialects. As a result, they achieved a 5% absolute decrease in the word error rate, from 56.8% to 51.5%.

#### D. Research Objective and Gap Analysis

Although the studies mentioned above have established a strong foundation for analyzing Arabic and Jordanian dialects, a significant gap remains in the granularity of dialect analysis. As summarized in **Table I**, most current research has treated the Jordanian dialect as a homogeneous unit or part of a broader regional area, or divided it into general sociolinguistic categories—urban, rural, and Bedouin—without considering specific geographical differences.

This study attempts to overcome this limitation through an in-depth research across three key Jordanian localities. By isolating these regions, we hope to discern particular phonological friction points and lexical variations that lead to recognition mistakes. Our aim is to shift from generalized Jordanian models toward a geographically aware framework that respects the significant linguistic diversity in the Kingdom. In this way, we hope to improve the recognition of users' "authentic voice" across all surveyed localities.

## IV. APPROACH AND METHODOLOGY

### A. Data Collection and Corpus Description

Textual prompts used are found in the Tatoeba corpus. Tatoeba is a free collection of example sentences with translations geared towards foreign language learners. Arabic sentences were chosen to collect dialectal speech data. The retrieved texts for the database span a variety of daily-life and conversational domains, touching on such topics as family relationships, work, transportation, education, travel, emotions, and social interaction. Speech data were collected from native speakers in three different regions of Jordan: Irbid, Amman, and Northern Aghwar. For each region, nearly **1500 samples** will be recorded, resulting in three parallel dialectal versions of the same textual material. Speakers are instructed to read the individual sentences in their natural local dialect. The dataset consists of male and female speakers with varying ages to guarantee diversity of speakers, increasing the generalization ability of models.

1) *Dataset Distribution:* We analyzed the distribution of sentence lengths to analyze the characteristics of the dataset. The distribution of the transcript lengths in the dataset is shown in Figure 1.

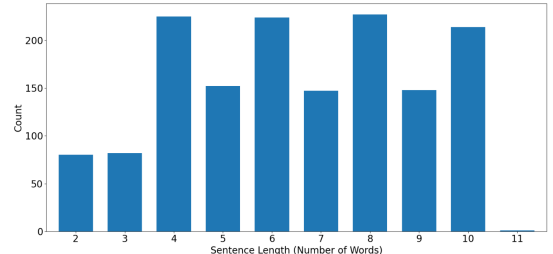


Figure 1: Distribution of sentence lengths in the dataset.

The lengths of the sentences range from 2 to 11 words, predominantly 4 to 10 words. The distribution across the various length categories is fairly even, with very short and very long sentences being less common. This diversity of sentence length provides a variety of speech patterns and linguistic structures, which is useful for training and evaluating the ASR models on different utterance conditions.

2) *Translation using GPT 5.2:* In the first stage of the preprocessing, after selecting appropriate sentences and localizing the data (such as Arabic names of people and places), the Arabic data was translated into the three dialects using GPT 5.2. We tested both Zero-shot and Few-shot as shown in the following **Table II**

Table I: Summary of Major Related Work on Arabic ASR and Dialectal Processing

| Year | Focus                    | Dataset/ Domain      | Model/ Method          | Main Contribution                                                        |
|------|--------------------------|----------------------|------------------------|--------------------------------------------------------------------------|
| 2017 | MSA ASR                  | Broadcast News       | GMM-HMM + DNN          | Improved acoustic modeling for MSA morphology.                           |
| 2018 | Dialect Identification   | MGB-3 (5 Dialects)   | CNN + Siamese          | Established baseline for regional dialect classification.                |
| 2019 | Multidialectal ASR       | MGB-5 (2,000h)       | Hybrid DNN-HMM         | Proved that multidialectal training improves cross-dialectal robustness. |
| 2021 | SSL for Low-Resource     | Mozilla Common Voice | Wav2Vec 2.0 (XLSR- 53) | Proved effectiveness of multilingual SSL for Arabic.                     |
| 2021 | Universal Repr           | 28 Languages         | XLS-R (Meta AI)        | Massive scaling of cross-lingual representations for low-resource ASR    |
| 2021 | Data Augmentation        | SASSC Corpus         | Noise/Speed/Pitch      | Achieved 4.55% WER reduction via training diversity.                     |
| 2021 | Multilingual/Code switch | QASR, MGB-3/5, CS    | Conformer + BPE        | Unified model handling dialectal and cross lingual switching             |
| 2022 | Jordanian Dialect ASR    | 113h Jordan Corpus   | Wav2Vec 2.0 + SSL      | Effective use of pseudo labeling for Jordanian Arabic.                   |
| 2024 | Arabic-Specific Arch.    | Varied MSA Corpora   | DeepSpeech2 + GRU      | Optimized inference and accuracy for Arabic phonetics.                   |
| 2024 | Multidialectal Bench.    | Casablanca Dataset   | Whisper-Mixed          | Benchmarked 8 dialects; showed mixed model superiority in generalization |
| 2025 | Generative Correction    | CoVoGER Benchmark    | LLM-based GER          | Used LLMs to contextually fix ASR errors across 15 languages.            |
| 2025 | Unified MSA/Classical    | MASC, FLEURS         | FastConformer          | Achieved state-of-the-art 1.34% WER for Classical Arabic                 |

Table II: Comparison of Results from Zero-shot and Few-shot Approaches

| MSA           | Zero Shot  | Few Shot      |
|---------------|------------|---------------|
| هاهي قوتك     | هاي أكلتك  | هاي قوتك      |
| تبدو لي قبيحة | شكل لي بشع | بتبين لي بشعة |
| رائحة كريهة   | ريحة بتقرف | كريهه ريحة    |

After testing both methods, we observed that the Zero-shot results were clearer and closer to the local pronunciation of Jordanian dialects, demonstrating a greater ability

to adapt to the local context flexibly, without the additional constraints imposed by the specific examples in Few-shot. This approach allowed the model to handle natural language more effectively, without any constraints on the model’s creativity or adherence to a specific pattern.

3) *Manual Review and Alignment with Guidelines* : After converting texts using GPT 5.2 MSA to local dialects, a manual review was conducted to ensure translation quality. Accordingly, the translation process followed a set of predefined guidelines specifically designed to account for the cultural and linguistic characteristics of Jordanian dialects, in addition to guidelines inspired by Obeidat et



al. [22] as shown in the following **Table III**. The accuracy of the dialect was found to be the most important factor in achieving the best translation results. [22]

Table III: Translation Guidelines for Dialectal Adaptation

| Guideline                                | Description                                                                                                                                                                                                                                                                         |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling the sound "تش"</b>           | Keep the sound tesh(تش) as it is and represent it using the letter kaf(ك)                                                                                                                                                                                                           |
| <b>Handling the letter qaf "ق"</b>       | For words containing the letter qaf(ق) we keep the original letter as it is, even if it is pronounced differently in some dialects                                                                                                                                                  |
| <b>Handling of Hamza "ء"</b>             | If the word has a commonly accepted written form without hamza (e.g., "نايم" written as "نايم"), we follow the common usage and remove the hamza. If the variation is from dialect pronunciation (e.g., "الجزائر" pronounced as "الجزائر"), we retain the original form with hamza. |
| <b>Length Constraints</b>                | The translation should stay close in length to the original sentence.                                                                                                                                                                                                               |
| <b>Fidelity to Source</b>                | The original meaning should be preserved without unnecessary changes.                                                                                                                                                                                                               |
| <b>Preserving Named Entities</b>         | Names of people, places, and brands should remain unchanged.                                                                                                                                                                                                                        |
| <b>Maintain Formatting</b>               | Keep punctuation, numbering, and overall structure as in the source.                                                                                                                                                                                                                |
| <b>Dialect Focus Constraint</b>          | Use expressions and patterns typical of the target dialect.                                                                                                                                                                                                                         |
| <b>Annotation Consistency Constraint</b> | Maintain consistent use of numbers, punctuation, and meaning throughout.                                                                                                                                                                                                            |

4) *Text and Audio Data Preprocessing*: The audio files were converted to a unified format (WAV, mono channel, 16 kHz sampling rate) to ensure compatibility with every selected ASR model. To improve audio quality, background noise was reduced using a noise sample taken from the start of each recording and leading and trailing silences were removed. Additionally, a Voice Activity Detection (VAD) system was used to detect and segment the speech portion of the audio files, ensuring that only the relevant speech segments were processed. Before normalization, textual data was manually adapted from the Modern Standard Arabic (MSA) into Jordanian dialectal forms to better reflect the natural spoken language. For each sentence, a dialectal version was created on a separate line

while preserving the original semantic content, resulting in parallel text entries aligned by unique sentence IDs as shown in the following **Table IV**.

Additionally, textual data were normalized to reduce spelling inconsistencies by eliminating diacritics, punctuation, and numbers and unifying common Arabic letter variants. This results in cleaner and standardized data for training and evaluation as shown in the following **Table V**.

Table V: Examples of Text Normalization

| Befor Normalization                | After Normalization              |
|------------------------------------|----------------------------------|
| جو الجبال يتغير بسرعة<br>إنته لإنه | انتبه لانه جو الجبال يتغير بسرعة |
| اوخذ بس ثنين شكرا<br>ماشي رح       | ماشي رح أوخذ بس ثنين شكراً,      |

Each audio file was matched with its corresponding dialect text using unique sentence IDs. Samples with missing or corrupted audio were removed. Random samples were manually checked to confirm the accuracy of audio-text alignment. The final dataset consisted of approximately 4,500 speech recordings, corresponding to nearly 1.9 hours. The dataset was divided into training, validation, and testing subsets using a 70%, 15%, and 15% split, respectively. The training set was used for model fine-tuning, while the validation set was used for hyperparameter selection. The test set was reserved for final evaluation on unseen data.

### B. speech recognition models

In this study, we evaluated several automatic speech recognition (ASR) models of varying sizes and structures to assess their ability to handle different Jordanian dialects, compare their performance, and determine the best model for the Jordanian dialect. The models included Four versions of Whisper: (Whisper Tiny, Whisper Small, Whisper Medium, and Whisper Large) as shown in the following **Table VI**. in addition to the Arabic model jonatasgrosman/wav2vec2-large-xlsr-53-arabic.

- **Whisper Models**: Whisper is a multilingual automatic speech recognition (ASR) model developed by OpenAI. It was pre-trained on a massive data set of 680,000 hours of speech, including more than 700 hours of Arabic data [?]. Whisper is based on a transformer encoder-decoder architecture. The process begins by converting the input audio into a log-Mel spectrogram. The encoder then extracts the important phonetic features from the signal, while the decoder gradually generates the corresponding text based on the extracted acoustic representation [?].

Table IV: Example of MSA to Jordanian Dialect Text Adaptation

| ID  | MSA                      | Irbid Dialect       | Amman Dialect      | NorthernAghwar Dialect |
|-----|--------------------------|---------------------|--------------------|------------------------|
| 132 | لماذا تنظرُ إليَّ هكذا ؟ | لويش بتلد علي هيك ؟ | ليش بتطلع في هيك ؟ | ليش بتطلع علي هيك ؟    |

Table VI: Architecture Details of the Whisper Model Variants

| Model          | Layers | Width | Heads | Parameters |
|----------------|--------|-------|-------|------------|
| Whisper Tiny   | 4      | 384   | 6     | 39M        |
| Whisper Small  | 12     | 768   | 12    | 244M       |
| Whisper Medium | 24     | 1024  | 16    | 769M       |
| Whisper Large  | 32     | 1280  | 20    | 1550M      |

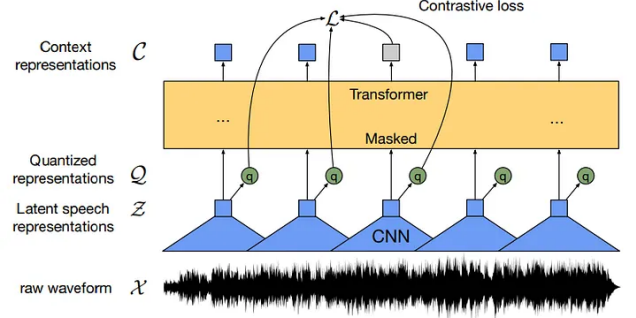


Figure 3: Architecture of the Wav2Vec 2.0 self-supervised speech representation model [23]

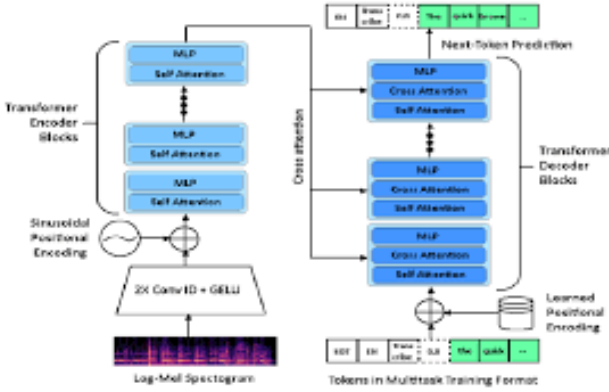


Figure 2: general structure [?]

- **Wav2Vec 2.0 Model:** The jonatasgrosmann/wav2vec2-large-xlsr-53-arabic model, based on the Wav2Vec 2.0 architecture (a self-learning framework developed by Meta AI for learning speech representation from large amounts of unlabeled audio data ,was evaluated. The selected model was pre-trained and fine-tuned specifically for Arabic speech using the XLSR-53 multilingual architecture, which is designed to handle linguistic variation and low-resource dialects [23].

### C. Cross-Dialect translation Strategy

After training each of the previous models on each dialect separately, the best model for each dialect was selected based on the lowest WER and CER values. Following the selection of the best model for each dialect

, it was tested on other Jordanian dialects to determine whether the model that learned the characteristics of a specific dialect could understand other Jordanian dialects, or whether its performance weakened when applied to a different dialect. This evaluation provides insight into the degree of similarity and linguistic overlap between Jordanian dialects. If the WER is low, it means that the similarity and convergence between dialects is high, while if it is high, there are differences in tone and sound. The goal of dialect cross-dialect evaluation is to measure the ability of models trained on a specific dialect to recognize other Jordanian dialects in order to study the degree of similarity and difference.

### D. Methodological Challenges and solutions

The processing of Arabic language faces numerous challenges due to the wide variety of dialects and pronunciation differences between different regions within the same country . These challenges require solutions to ensure consistency in the speech-to-text process. In this context , we present some of the solutions we have adopted to address these challenges:

- **Dialectal Variation and Formal Transcription Challenges :**The differences between spoken dialects and Standard Arabic make it difficult to choose between phonetic and formal transcription. To address this, we prioritized a unified transcription of the language , based on the guidelines provided in the **Table III**,while ensuring that the model is exposed to a variety of pronunciations.
- **Spelling Inconsistencies Due to Pronunciation Variations :**The variation in pronunciation between dialects



leads to inconsistencies in spelling. To address this, we adopted standardized spelling while allowing speakers to use their usual natural pronunciation during recording to reflect the reality that people's pronunciation sometimes differs from their spelling.

- **Handling the sound "ك"** :The letter "ك" (k), which sounds like kaf "كاف", is pronounced like the sound "تش" (e.g., "كلب" pronounced as "تشلب"). To ensure consistency in the data, we retained the letter "ك" (k) in the written data while allowing all sounds in the audio data, ensuring that the pronunciation is natural, as pronounced by the speaker.
- **Handling of Hamza "ء"** :Some words are written and pronounced without the hamza uniformly throughout Jordan, such as "نائم", which is written and pronounced "نايم". Other words have a different spelling than their common pronunciation, such as "الجزائر", which is written "الجزائر" but pronounced "الجزاير" in some Jordanian dialects. Therefore, the hamza is dropped when spelling and pronunciation are locally standardized and agreed on, while it is retained in words that are usually written in Standard Arabic.

- **Data Preprocessing** :Because the data we used was foreign, found on the Tatoeba corpus., we manually cleaned it to ensure its suitability for the Arab community. This preprocessing involved excluding sentences that might be socially or morally inappropriate, with the aim of ensuring the model was not exposed to content that was unsuitable or culturally objectionable in the Arab context.

## V. EXPERIMENTAL SETUP

### A. Dataset Splitting

A split ratio of 70% for training, 15% for validation, and 15% for testing was adopted. The training set was used for model training, hyperparameters were tuned and the model selected on the validation set, and the test set was held out for the final evaluation.

### B. Evaluation Metrics

To improve the performance of these ASR models, WER and CER were used as the main evaluation metrics.

**WER** measures the percentage of word-level errors between the predicted transcription and the reference text. It is widely used as a standard measure in speech recognition experiments. The formula for WER is:

$$\text{WER} = \frac{S + D + I}{N} \times 100 \quad (1)$$

Where:  $S$  is the number of substitutions, (2)

$D$  is the number of deletions, (3)

$I$  is the number of insertions, (4)

$N$  is the total number of words in the reference text. (5)

**CER** was used to detect character-level errors, giving a more detailed evaluation of the model's performance. This is important for Arabic dialects due to the numerous pronunciation and spelling differences. The formula for CER is:

$$\text{CER} = \frac{S + D + I}{N} \times 100 \quad (6)$$

Where:  $S$  :the number of substitutions at the character level (7)

$D$  :the number of deletions at the character level (8)

$I$  :the number of insertions at the character level (9)

$N$  :the total number of characters in the reference text (10)

The use of both WER and CER covers both word-level errors and character-level micro-errors.

### C. Hyperparameter Tuning

To determine the optimal training settings for each model, several hyperparameter combinations were evaluated. The selected search space was based on commonly used settings for fine-tuning transformer-based automatic speech recognition systems and on preliminary experiments conducted on the Jordanian dialect dataset. Low learning rates were adopted to ensure stable adaptation of the pre-trained models, while different batch sizes were evaluated to balance training stability and available computational resources. AdamW and Adafactor were selected due to their effectiveness in training large transformer-based architectures. **Table VII** summarizes the evaluated hyperparameters.

Table VII: Hyperparameter Search Space

| Hyperparameter   | Values           |
|------------------|------------------|
| Learning Rate    | 1e-05, 2e-05     |
| Batch Size       | 4, 8             |
| Optimizer        | AdamW, Adafactor |
| Number of Epochs | 5                |

The performance of each configuration was monitored on the validation set using WER and CER. The configuration achieving the lowest validation WER was selected

as the final model configuration for each dialect and was subsequently used in the cross-dialect evaluation experiments.

## VI. RESULTS AND DISCUSSION

### A. Model Performance and Best Configuration Selection

In this study, we did not limit our evaluation to testing the models directly. Instead, our primary objective was to investigate whether fine-tuning on local Jordanian dialect data could improve speech recognition performance.

To achieve this, the performance of each model was measured before and after fine-tuning. The first set of experiments was conducted on the Irbid dialect dataset using Whisper Tiny, Whisper Small, Whisper Medium, and Whisper Large. Before fine-tuning (General-purpose model), the models exhibited relatively high error rates, which is expected since they were pre-trained on general speech data and were not specifically adapted to the Irbid dialect. After fine-tuning (Dialect-aware mode), all model variants showed substantial improvements in both WER and CER.

**Table VIII** summarizes the performance of Automatic Speech Recognition (ASR) models on the Irbid dialect dataset before and after fine-tuning. All Whisper versions showed significant and noticeable improvements after fine-tuning.

A clear relationship was also found between model size and performance. Larger models achieved lower WER and CER. The Whisper Large model achieved the best overall performance with a WER of (23.8%) and a CER of (7.9%), followed by the Whisper Medium model with a WER of (26.3%) and a CER of (8.5%). Therefore, the results indicate that larger models are better able to capture dialect-specific acoustic and linguistic characteristics.

Tables IX and X present the performance of the ASR models evaluated in the data sets of the northern Aghwar and Amman dialects before and after fine-tuning. The same pattern emerged: larger models yielded lower WER and CER values. Whisper Large performed best again, supporting its selection for subsequent cross-dialect evaluation experiments.

When analyzing the results obtained from the three dialects, it can be seen that the Amman dialect achieved the lowest values of WER(14.7%) and CER(4.46%) after the fine-tuning process, while the Irbid dialect recorded the highest values of WER(23.8%) and CER(7.9%).

The Amman dialect performed better, possibly due to its closer resemblance to the urban dialect, which is commonly spoken in Jordan. This dialect may also be more similar to the Arabic speech patterns that the model encountered during the pre-training stage. Since Whisper was trained on large amounts of audio data taken from the web, including media, conversational, and general content aimed at a broad audience [?], the speech patterns represented in these data are likely closer to urban dialects

and widespread language patterns than to more specific local dialects.

The highest error rates were recorded in the Irbid dialect, which may be attributed to the greater diversity of local speech patterns within this dialect. This increased variability may reduce the model's ability to learn a consistent representation of the dialect.

However, the Northern Aghwar dialect achieved intermediate performance between Irbid and Amman, and this may indicate that the Northern Aghwar dialect has a higher degree of linguistic and phonetic homogeneity compared to the Irbid dialect, making it easier for the model to learn its characteristics and adapt to them more effectively.

### B. Cross-Dialect translation Results

After comparing the performance of the evaluated models and selecting the best-performing model for each dialect, cross-dialect evaluation trials were conducted to study the ability of the Automatic Speech Recognition (ASR) models for each dialect to generalize to other Jordanian dialects.

Table XI: Cross-Dialect Evaluation Results

| Training Dialect | Evaluation Dialect | WER (%) | CER (%) |
|------------------|--------------------|---------|---------|
| Northern Aghwar  | Northern Aghwar    | 15.6    | 4       |
| Northern Aghwar  | Irbid              | 28.59   | 9.33    |
| Northern Aghwar  | Amman              | 20.14   | 6.90    |
| Irbid            | Irbid              | 23.8    | 7.9     |
| Irbid            | Northern Aghwar    | 16.38   | 5.60    |
| Irbid            | Amman              | 18.60   | 5.48    |
| Amman            | Amman              | 14.70   | 4.46    |
| Amman            | Irbid              | 45.26   | 24.89   |
| Amman            | Northern Aghwar    | 20.8    | 5.99    |

Table XI presents the results of these trials. The Training Dialect column indicates the dialect on which the model was trained, while the Evaluation Dialect column represents the dialect used for testing.

The results indicate that the Irbid dialect was the most difficult and distinctive for the model. The model trained on the Irbid dialect achieved the best performance when tested on the same dialect, with a WER value of approximately 23.8%. However, training on the Irbid dialect gave the model good generalizability to other dialects. When tested on the Aghwar dialect, the Irbid-trained model achieved a WER of 16.38%, a result very close to the best result for Aghwar, which is 15.6%, with a difference of only 0.78%. When tested on the Amman dialect, it achieved a WER of 18.60%, which is relatively close to the best result for Amman, which is 14.70%, with a difference of 3.90%.

Therefore, the results indicate that the Irbid dialect, despite its difficulty and complexity during training, contains a wide range of phonetic and linguistic diversity, giving

Table VIII: Performance of models in the Irbid Dialect : before and after fine-tuning

| Model                  | WER Before (%) | WER After (%) | CER Before (%) | CER After (%) | Learning Rate | Batch Size | Optimizer |
|------------------------|----------------|---------------|----------------|---------------|---------------|------------|-----------|
| Whisper Tiny           | 140.0          | 66.9          | 107.0          | 26.12         | 2e-05         | 8          | Adafactor |
| Whisper Small          | 85.0           | 38.81         | 46.0           | 12.88         | 2e-05         | 8          | Adafactor |
| Whisper Medium         | 78.3           | 26.3          | 27.1           | 8.5           | 1e-05         | 8          | AdamW     |
| Whisper Large          | 67.0           | 23.8          | 30.8           | 7.9           | 1e-05         | 8          | AdamW     |
| Jonatas XLSR-53 Arabic | 91.51          | 71.91         | 37.52          | 45.18         | 1e-05         | 8          | AdamW     |

Table IX: Performance of models in the Northern Aghwar Dialect : before and after fine-tuning

| Model                  | WER Before (%) | WER After (%) | CER Before (%) | CER After (%) | Learning Rate | Batch Size | Optimizer |
|------------------------|----------------|---------------|----------------|---------------|---------------|------------|-----------|
| Whisper Tiny           | 119.0          | 53.44         | 77.0           | 18.08         | 2e-05         | 4          | AdamW     |
| Whisper Small          | 67.7           | 25.31         | 29.1           | 6.85          | 2e-05         | 4          | AdamW     |
| Whisper Medium         | 60.3           | 16.97         | 28.75          | 4.2           | 1e-05         | 8          | Adafactor |
| Whisper Large          | 57.7           | 15.6          | 25.9           | 4.0           | 1e-05         | 8          | AdamW     |
| Jonatas XLSR-53 Arabic | 85.15          | 48.85         | 37.52          | 13.23         | 2e-05         | 4          | Adafactor |

Table X: Performance of models in the Amman Dialect : before and after fine-tuning

| Model                  | WER Before (%) | WER After (%) | CER Before (%) | CER After (%) | Learning Rate | Batch Size | Optimizer |
|------------------------|----------------|---------------|----------------|---------------|---------------|------------|-----------|
| Whisper Tiny           | 96.42          | 49.04         | 61.44          | 16.71         | 2e-05         | 8          | Adafactor |
| Whisper Small          | 62.21          | 23.33         | 24.79          | 7.01          | 2e-05         | 4          | Adafactor |
| Whisper Medium         | 60.12          | 17.47         | 25.87          | 5.16          | 2e-05         | 8          | AdamW     |
| Whisper Large          | 50.18          | 14.70         | 66.85          | 4.46          | 2e-05         | 8          | AdamW     |
| Jonatas XLSR-53 Arabic | 85             | 53.12         | 35.52          | 15.04         | 2e-05         | 8          | AdamW     |

the model exposure to more varied phonetic patterns and helping it to generalize better to other dialects.

The Amman dialect was the easiest for the model to train and test on, achieving the best WER of 14.70%. However, the model trained on Amman did not generalize well to the Irbid dialect, where the WER value increased to 45.26%, the highest error rate among all results. This indicates that the Amman dialect may be less diverse and closer to the general pattern, and therefore does not cover the phonetic characteristics of the Irbid dialect.

As for the Northern Aghwar dialect, it showed similar behavior to the Amman dialect, as it did not generalize strongly to the Irbid dialect, with a WER of 28.59%. However, it performed acceptably on the Amman dialect at 20.14%. In contrast, the Amman model gave a very close result on the Northern Aghwar dialect at 20.8%. The difference between these two cross-dialect results is only 0.66% , indicating a very clear convergence between the phonetic patterns of the Amman and Northern Aghwar dialects compared to the Irbid dialect.

In general, automatic speech recognition (ASR) models were evaluated based on their ability to generalize across different Jordanian dialects. The results strongly indicated that the ASR model trained on the Irbid dialect demonstrated the best generalization ability with relatively low error rates when evaluated on dialects outside its training domain, such as the Northern Aghwar dialect and the Amman dialect. Conversely, the performance of the models varied according to the degree of similarity

between the trained dialect and the tested dialect, indicating that the model’s generalization ability is closely related to shared phonological and linguistic characteristics. Therefore, from the perspective of ASSR systems, these performance differences provided a practical, data-driven indicator of the phonological and structural similarities between Jordanian dialects.

### C. Qualitative Error Analysis

In addition to quantitative assessments using WER and CER, a qualitative analysis was conducted to study the types of errors produced by the ASR models that were evaluated.

The tables XII, XIII, and XIV presents selected examples from experiments conducted on the three dialects. These results show that most errors were not complete failures in sentence or word recognition. In most cases, the model retained the general meaning of the sentence despite minor spelling errors, and most of the errors were word substitutions.

The predicted text remained semantically similar to the reference sentence despite the recognition errors. These observations help explain the relatively low matching error rates obtained in many experiments. Despite partially incorrect word recognition, the models often preserved the letter sequence, sentence structure, and meaning, indicating that the extracted phonetic representations successfully captured important dialectal features.

Table XII: Qualitative Error Analysis Examples for the Irbid Dataset

| Ground Truth            | Whisper IRBID           | Northern Aghwar Model  | Amman Model              |
|-------------------------|-------------------------|------------------------|--------------------------|
| حكائنا سامي قصه حياته   | حكائنا سامي دصه بحياته  | حكائنا سامي سفتح حياته | حكائنا سامي قصح حياته    |
| معمروهوش سامي اجاع داري | معمروهوش سامي اجاع داري | معمروهوش سامي اجاعداني | معمره سامي اجاع علي بيتي |
| قديش عمره ابك           | قديش عمرو ابك           | قديش عمره بك           | كديش عمر بنك             |

Table XIII: Qualitative Error Analysis Examples for the Al-Aghwar Dataset

| Ground Truth            | Whisper Northern Aghwar | Irbid Model               | Amman Model            |
|-------------------------|-------------------------|---------------------------|------------------------|
| قالنا سامي قصه حياته    | قالنا سامي قصه حياتك    | قالنا سامي قصه حياه       | قالنا سامي قصه حياك    |
| بعمرو سامي ما اجاع داري | بعمره سامي ما اجاع داري | بعمره سامي ما اجعه ع داري | بعمره سامي ما اجم داري |
| قديش عمر ابك            | قديش عمرو ابك           | قديش عمر ابك              | كداش عمر ابك           |

Table XIV: Qualitative Error Analysis Examples for the Amman Dataset

| Ground Truth               | Whisper Amman              | Irbid Model                | Northern Aghwar Model      |
|----------------------------|----------------------------|----------------------------|----------------------------|
| حكائنا سامي قصه حياته      | حكائنا سامي قصه حياته      | حكائنا سامي قصه حياته      | حكائنا سامي قصه حياته      |
| ما عمره سامي اجاع علي بيتي | ما عمره سامي اجاع علي بيتي | ما عمره سامي اجاع علي بيتي | ما عمره سامي اجاع علي بيتي |
| قديش عمر ابك               | كديش عمر ابك               | قديش عمر ابك               | ديش عمر ابك                |

## VII. CONCLUSION

In this study, we worked on building a dialectal speech corpus for Jordanian dialects and simultaneously trained and evaluated ASR models for Jordanian dialectal speech. We focused on three regions (Irbid, Amman, and Northern Aghwa). The goal was not to treat Jordanian dialects as a single entity or dialect, but rather to study the differences between dialects according to region. The first step was collecting and preparing data, adapting Arabic sentences into local Jordanian dialects based on standardized transcription and dialectal adaptation guidelines, and then recording them in audio format by native speakers from these regions. 4500 audio recordings were collected, including 1.9 hours of speech. Then, we built a pre-processing setup: standardizing the audio to WAV, mono, and 16 kHz, with noise reduction, silence trimming, and VAD. The texts were also cleaned, normalized, spelling inconsistencies and transcription differences were reduced. We tested more than one ASR model and identified different versions of Whisper: Tiny, Small, Medium, and Large, in addition to the Arabic Wav2Vec 2.0 model. The goal was to compare the models before and after fine-tuning with Jordanian dialects. We used WER and CER as evaluation measures, so we measured word-level and character-level errors. The results showed that Fine-tuning significantly improved ASR performance general-purpose models before fine-tuning. The pattern also showed that each time, the Whisper model was the most frequently used, so Whisper Large was the best overall and achieved better results after training with the three dialects. For example, in Amman, it reached WER 14.70%, in Northern Aghwar 15.%, and in Irbid 23.8%.

We conducted cross-dialect evaluation, meaning we trained the model on a specific dialect and tested it on other dialects to determine if the model could generalize

to other dialects. Here, it became clear that the Irbid dialect was the most difficult and distinctive, because the models trained in Amman or Northern Aghwar exhibited errors when tested in Irbid, especially the Amman model, which reached 45.26% WER in Irbid dialect.

In contrast, the model trained in Irbid performed better in Northern Aghwar and Amman, achieving 16.38% WER in Northern Aghwar and 18.60% WER in Amman. The important point here is that although Irbid is more challenging and complex for the models, training on it exposes the model to a wider range of phonetic and linguistic patterns, thus helping them generalize to other dialects. As for Amman and Northern Aghwar, a specific phonetic type became known because the cross-dialect results were very close, with a difference of about 0.66%. Furthermore, qualitative error analysis was also conducted, meaning we don't just rely on numbers, but also examine examples of the errors. In many cases, the failure wasn't complete, often, the errors were often substitutions or spelling/pronunciation variations, yet the overall meaning of the sentence remained close to the correct text. This supports the idea that the models were able to learn important dialectal speech features, even if the transcriptions aren't perfect.

## REFERENCES

- [1] B. Talafha, K. Kadaoui, S. M. Magdy, M. Habiboullah, C. M. Chafei, A. O. El-Shangiti, H. Zayed, M. C. Tourad, R. Alhamouri, R. Assi, A. Alraesi, H. Mohamed, F. Alwajih, A. Mohamed, A. El Mekki, E. M. B. Nagoudi, B. D. M. Saadia, H. A. Alsayadi, W. Al-Dhabyani, S. Shatnawi, Y. Ech-chammakhy, A. Makouar, Y. Berrachedi, M. Jarrar, S. Shehata, I. Berrada, and M. Abdul-Mageed, "Casablanca: Data and models for multidialectal Arabic speech recognition," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Y. Al-Onaizan, M. Bansal, and Y.-N. Chen, Eds. Miami, Florida, USA: Association for Computational

- Linguistics, Nov. 2024, pp. 21 745–21 758. [Online]. Available: <https://aclanthology.org/2024.emnlp-main.1211/>
- [2] K. Abu Kwaik, M. Saad, S. Chatzikyriakidis, and S. Dobnik, “Shami: A corpus of levantine arabic dialects,” in *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*. European Language Resources Association (ELRA), 2018, pp. 1–6. [Online]. Available: <https://aclanthology.org/L18-1576/>
  - [3] UNESCO, “Arabic language, a bridge between civilizations,” <https://unesdoc.unesco.org/ark:/48223/pf0000380347>, 2021, accessed: 2026-01-03.
  - [4] A. Dhouib, A. Othman, O. El Ghoul, M. K. Khribi, and A. Al Sinani, “Arabic automatic speech recognition: A systematic literature review,” *Applied Sciences*, vol. 12, no. 17, p. 8898, 2022.
  - [5] E.-S. Badawi, “Levels of contemporary arabic in egypt,” 1996.
  - [6] J. Dunn and S. Wong, “Stability of syntactic dialect classification over space and time,” in *Proceedings of the 29th International Conference on Computational Linguistics*, N. Calzolari, C.-R. Huang, H. Kim, J. Pustejovsky, L. Wanner, K.-S. Choi, P.-M. Ryu, H.-H. Chen, L. Donatelli, H. Ji, S. Kurohashi, P. Paggio, N. Xue, S. Kim, Y. Hahm, Z. He, T. K. Lee, E. Santus, F. Bond, and S.-H. Na, Eds. Gyeongju, Republic of Korea: International Committee on Computational Linguistics, Oct. 2022, pp. 26–36. [Online]. Available: <https://aclanthology.org/2022.coling-1.3/>
  - [7] Q. Li, Q. Mai, M. Wang *et al.*, “Chinese dialect speech recognition: a comprehensive survey,” *Artificial Intelligence Review*, vol. 57, no. 25, 2024. [Online]. Available: <https://doi.org/10.1007/s10462-023-10668-0>
  - [8] J. Lee, K. Kim, and M. Chung, “Korean dialect identification based on intonation modeling,” in *2021 24th Conference of the Oriental COCOSDA International Committee for the Coordination and Standardisation of Speech Databases and Assessment Techniques (O-COCOSDA)*. IEEE, 2021, pp. 168–173.
  - [9] D. Alves, “Information theory and linguistic variation: A study of Brazilian and European Portuguese,” in *Proceedings of the 12th Workshop on NLP for Similar Languages, Varieties and Dialects*, Y. Scherrer, T. Jauhainen, N. Ljubešić, P. Nakov, J. Tiedemann, and M. Zampieri, Eds. Abu Dhabi, UAE: Association for Computational Linguistics, Jan. 2025, pp. 9–19. [Online]. Available: <https://aclanthology.org/2025.vardial-1.2/>
  - [10] N. Sengupta, S. K. Sahu, B. Jia, S. Katipomu, H. Li, F. Koto, W. Marshall, G. Gosal, C. Liu, Z. Chen, O. M. Afzal, S. Kamboj, O. Pandit, R. Pal, L. Pradhan, Z. M. Mujahid, M. Baali, X. Han, S. M. Bsharat, A. F. Aji, Z. Shen, Z. Liu, N. Vassilieva, J. Hestness, A. Hock, A. Feldman, J. Lee, A. Jackson, H. X. Ren, P. Nakov, T. Baldwin, and E. Xing, “Jais and jais-chat: Arabic-centric foundation and instruction-tuned open generative large language models,” 2023. [Online]. Available: <https://arxiv.org/abs/2308.16149>
  - [11] M. A. Menacer, O. Mella, D. Fohr, D. Jouvet, D. Langlois, and K. Smaili, “An enhanced automatic speech recognition system for Arabic,” in *Proceedings of the Third Arabic Natural Language Processing Workshop*, N. Habash, M. Diab, K. Darwish, W. El-Hajj, H. Al-Khalifa, H. Bouamor, N. Tomeh, M. El-Hajj, and W. Zaghouani, Eds. Valencia, Spain: Association for Computational Linguistics, Apr. 2017, pp. 157–165. [Online]. Available: <https://aclanthology.org/W17-1319/>
  - [12] H. Alsayadi, A. Abdelhamid, I. Hegazy, and Z. Taha, “Data augmentation for arabic speech recognition based on end-to-end deep learning,” *International Journal of Intelligent Computing and Information Sciences*, vol. 21, no. 2, pp. 50–64, 2021.
  - [13] M. M. Nasef, A. A. Elshall, and A. M. Sauber, “Arabrecog-nizer: modern standard arabic speech recognition inspired by deepspeech2 utilizing franco-arabic,” *International Journal of Speech Technology*, vol. 27, no. 3, pp. 673–686, 2024.
  - [14] T. Zouhair, “Automatic speech recognition for low-resource languages using wav2vec2: Modern standard arabic (msa) as an example of a low-resource language,” 2021.
  - [15] A. Babu, C. Wang, A. Tjandra, K. Gulati, F. Metze, and Y. Gu, “Xls-r: Self-supervised cross-lingual speech representation learning at scale,” *arXiv preprint arXiv:2111.09296*, 2021. [Online]. Available: <https://arxiv.org/abs/2111.09296>
  - [16] A. Ali, S. Shon, Y. Samih, H. Mubarak, A. Abdelali, J. Glass, S. Renals, and K. Choukri, “The mgb-5 challenge: Recognition and dialect identification of dialectal arabic speech,” in *2019 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, 2019, pp. 1026–1033.
  - [17] S. Shon, A. Ali, and J. Glass, “Convolutional neural networks and language embeddings for end-to-end dialect recognition,” in *Proceedings of Odyssey 2018: The Speaker and Language Recognition Workshop*, 2018, pp. 98–104. [Online]. Available: <https://arxiv.org/abs/1803.04567>
  - [18] S. A. Chowdhury, A. Hussein, A. Abdelali, and A. Ali, “Towards one model to rule all: Multilingual strategy for dialectal code-switching arabic ASR,” in *Proceedings of Interspeech 2021*, 2021, pp. 2466–2470. [Online]. Available: <https://arxiv.org/abs/2105.14779>
  - [19] L. Grigoryan, N. Karpov, E. Albasiri, V. Lavrukhin, and B. Ginsburg, “Open automatic speech recognition models for classical and modern standard arabic,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, arXiv:2507.13977. [Online]. Available: <https://arxiv.org/abs/2507.13977>
  - [20] Z. Yang, Z. Wan, S. Li, C.-H. H. Yang, and C. Chu, “CoVoGER: A multilingual multitask benchmark for speech-to-text generative error correction with large language models,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds. Suzhou, China: Association for Computational Linguistics, Nov. 2025, pp. 6313–6325. [Online]. Available: <https://aclanthology.org/2025.emnlp-main.320/>
  - [21] A. A. Safieh, I. A. Alhaol, and R. Ghnemat, “End-to-end jordanian dialect speech-to-text self-supervised learning framework,” *Frontiers in Robotics and AI*, vol. 9, p. 1090012, 2022. [Online]. Available: <https://doi.org/10.3389/frobt.2022.1090012>
  - [22] R. Obeidat, L. Alawneh, and Y. Al-Harashsheh, “Exploring prompting for dialectal machine translation: a focus on north jordanian arabic,” *PeerJ Comput. Sci.*, vol. 11, p. e3209, 2025. [Online]. Available: <https://api.semanticscholar.org/CorpusID:283007759>
  - [23] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” *Advances in neural information processing systems*, vol. 33, pp. 12 449–12 460, 2020.